PHYSICAL CHEMISTRY

GUIDED REVISION PLAN_SCORE ADVANCED [SOLUTION]

Topic : Atomic Structure (Question Bank # 12)

SECTION-I (I) ONLY ONE OPTION CORRECT TYPE

1. **(C)**

Sol. $\lambda v_{k_\alpha} \propto (Z-1)^2$ for k_λ - characteristic x-ray

$$\frac{hc}{\lambda_{k_\alpha}} \propto (Z-1)^2 \Rightarrow \frac{(\lambda_{k_\alpha})_2}{(\lambda_{k_\alpha})_1} = \frac{(Z_1-1)^2}{(Z_2-1)^2}$$

hence $4 = \left(\frac{Z_1-1}{Z_2-1}\right)^2$ as $\frac{Z_1-1}{Z_2-1} = 2$

$$\Rightarrow Z_2 = \frac{Z_1+1}{2}$$

2. **(C)**

Sol. For H-atom is ground state Z = 1, n = 1

$$\text{Frequency of revolution} = \frac{v}{2\pi r} = \frac{2.188\times10^6}{2\times3.142\times0.529\times10^{-10}}$$

= frequency of emitted radiation

wavelength (λ) of radiation $= \frac{c}{\nu}$

$$= \frac{3.0\times10^8\times2.188\times10^6}{2\times3.142\times0.529\times10^{-10}}$$

$\simeq 4.5\times10^{-8}$ m = 45nm

3. **(A)**

Sol. For H-atom angular momentum of $e^\ominus$

is given by $n\left(\frac{h}{2\pi}\right)$

on absorbing 10.2 eV radiation $e^\ominus$ jumps
from n = 1 (ground state) to n = 2

Hence change in angular momentum $= 2\left(\frac{h}{2\pi}\right) - 1\left(\frac{h}{2\pi}\right)$

Ans. (A) $= \frac{h}{2\pi}$

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4. **(D)**

Sol. 2 radial and 2 angular nodes $\Rightarrow$ 5d subshell
same sign of wavefunction in all direction $\Rightarrow$ 1s subshell
$\Delta E = Z^2 (13.6 - 0.54) = 326.4$
$\Rightarrow Z = 5$ Hence, it should be $B^{4\oplus}$
Ans. (D)

5. **(D)**

Sol. Energy requuired for dissociation of O_2 molecule into two normal O-atoms
$= 482.5$ kJ / mol

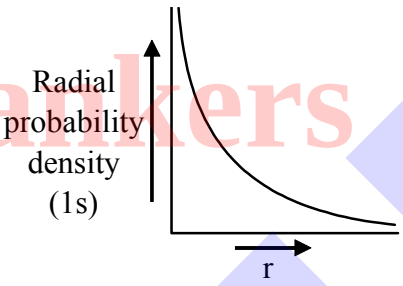
$$= \frac{482.5}{96.5} \text{eV / molecule}$$

$\simeq 5$eV / molecule
Since, one of the dissociated O-atom is 1.2 eV more energetic than normal
Energy required for the dissociation of one O_2 molecule $= 5 + 1.2 = 6.2$ eV

$$\lambda = \frac{hC}{\Delta E} = \frac{1242}{6.2} \simeq 200\text{nm}$$ Ans. (D)

6 **(C)**

Sol.



Radial probability density decreases with increase in 'r'
Hence, $P_1 = P_3 > P_2 = P_4$

7. **(D)**

Sol. The electronic level which would allow the hydrogen atom to absorb a photon but not to emit a photon should be the lowest energy level. hence, the Ans. is 1s
Ans. (D)

8. **(C)**

Sol. $$\lambda = \frac{h}{mv} = \frac{h}{p} \Rightarrow p = \frac{h}{\lambda} \Rightarrow dp = -\frac{h}{\lambda^2} d\lambda$$

$$dp.dx = \frac{h}{4\pi}$$

$$\frac{\lambda}{\lambda^2} d\lambda.dx = \frac{h}{4\pi} \Rightarrow d\lambda.dx = \frac{\lambda^2}{4\pi}$$

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$$\lambda = \sqrt{\frac{150}{v}}\text{Å} = \sqrt{\frac{150}{6}}\text{Å} = 5\text{Å}$$

$$d\lambda = \frac{\lambda^2}{4\pi.dx} = \frac{5\times5}{4\times\frac{22}{7}\times\frac{7}{22}\times10}\text{ Å} = \frac{5}{8}\text{Å}$$

$= 0.625\text{Å}$ Ans. (C)

9. (C)

Sol. $\frac{hc}{\lambda} = \phi + 1ev$ .....(1)

$\frac{3hc}{\lambda} = \phi + 4ev$ .....(2)

$(2)-(1) \Rightarrow \frac{2hc}{\lambda} = 3ev \quad \Rightarrow \frac{hc}{\lambda} = 1.5\,ev$

By (1) we can say $\phi = 0.5\,eV$

Ans. (C)

MULTIPLE CORRECT

10. (A,B)

Sol. Orbital angular momentum of an e^- in a single electro system $= \sqrt{\ell(\ell+1)}\frac{h}{2\pi} = \frac{\sqrt{3}h}{\pi}$

$\Rightarrow \quad \ell(\ell+1) = 3\times4$

$\Rightarrow \quad \ell = 3$

$\therefore$ possible values of $n = 4, 5, ...$

Hence possible values of angular momentum

$\frac{4h}{2\pi}, \frac{5h}{2\pi}, ...$

And not possible values are $\frac{h}{2\pi}, \frac{2h}{2\pi}, \frac{3h}{2\pi}$

Hence, Answer $\left(\frac{h}{\pi}\right)\left(\frac{3h}{2\pi}\right)$

11. (A,B,D)

Sol. For He^+ ion $r_3 - r_2 = (3^2-2^2)_{\frac{r_0}{z}=\frac{5r_0}{z}}$

(A) $r_4 - r_3 = (4^2-3^2)_{\frac{r_0}{z}=\frac{7r_0}{z}}$

$v \propto \frac{z}{h} \qquad \frac{5r_0}{z} < \frac{7r_0}{z}$

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(B) $\frac{v_3}{v_4} = \frac{4}{3}$ (greater than one)

(C) $E \propto \frac{z^2}{n^2}$ Hence, ΔE decreases continuously

(D) $\frac{\Delta E_{\text{leart energetic}}}{\Delta E_{\text{most energetic}}} = \frac{\frac{1}{1^2} - \frac{1}{2^2}}{\frac{1}{1^2} - \frac{1}{(\infty)^2}} = \frac{3}{4}$

Since $\Delta E \propto \frac{1}{\lambda}$, hence,

$\frac{\lambda_{\text{least energetic}}}{\lambda_{\text{most energetic}}} = \frac{4}{3}$

Hence, Ans is A,B,D

12. (B,D)

Sol.

orbitals	No. of spherical nodes
2s	1
4p	2
3d	0
6f	2

Ans. is 4p, 6f

13. (A,C)

Sol.

Orbitals	Radial nodes	Angular nodes
3p	1	1
5d	2	2

Ans is 3p, 5d,

Hence, (A,C)

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14. (A,B,C)

Sol. Metal A

$$E_{photon} = \phi_A + K.E_{max}$$

$$4.25 = \phi_A + \frac{p^2}{2m}$$

$$= \phi_A + \frac{h^2}{2m(2\lambda_A)^2}$$

$$\frac{h^2}{2m\lambda_A^2} - 1.5 = \frac{h^2}{2m(2\lambda_A)^2}$$

$$\frac{h^2}{2m\lambda_A^2} \times \frac{3}{4} = 1.5$$

Metal B

$$E_{photon} = \phi_B + K.E_{max}$$

$$4.7 = \phi_B + \left(\frac{h^2}{2m(\lambda_A)^2} - 1.5\right)$$

$$K.E._{max}(A) = 1.5 \times \frac{4}{3} = 2.0 \text{ eV}$$

$$K.E._{max}(B) = 2.0 - 1.5 = 0.5 \text{ eV}$$

$$\phi_A = 4.25 - 2.00 = 2.25 \text{ eV}$$

$$\phi_B = 4.7 - 0.5 = 4.2 \text{ eV}$$

Hence, Answer (A,B,C)

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PARAGRAPH

Paragraph (Q.15 to Q.17)

15. (B)

Sol. S_1 is symmetric and has one radial node $\Rightarrow$ 2s

S_2 has one radial node and its energy is equal to the energy of the ground state of H-atom $\Rightarrow n = 3$

$\therefore \quad S_2 = 3p$

16. (C)

Sol. $Es_1 = \frac{z^2}{n^2}(E_H) = \frac{3^2}{2^2}(E_H) = 2.25\ E_H$

17. (B)

Sol. Orbital angular momentum quantum number for $S_2(3p) \Rightarrow \ell = 1$

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Paragraph (Q.18 to Q.20)

18. (C)

Sol. $r = a_0$ = Bohr's orbit

$\rightarrow$ there is maximum probability of finding the e^-

$$\frac{d(4\pi r^2\psi_r^2)}{dr} = 0 \Rightarrow r = a_0$$

19. (A)

Sol. $\Psi_{310} = 3p_2$

radial function for $3p_y$ and $3p_z$ are

$$\psi_\lambda \text{ for } 3p_y = \frac{1}{9\sqrt{3}}\left(\frac{1}{a_0}\right)^{3/2}(4-\sigma)\,\sigma e^{-\sigma/2}$$

If $\psi_r = 0 \Rightarrow \sigma = 0, 4, \infty$

$\Rightarrow$ $\sigma = 4$ for spherical node

$$\Rightarrow \frac{2r}{na_0} = \frac{2r}{3a_0} = 4$$

$\Rightarrow$ $r = 6a_0$

Hence Answer is (A)

20. (D)

Sol. For 3p orbital-which has one radial node- the graph will intersect x-axis at one non-zero point.

SECTION-II (INTEGER SECOND DECIMAL)

1. (11.00)

Sol.

n = 1	1s			
n = 2	2s	2p		one maxima only
n = 3	3s	3p	3d	
n = 4	4s	4p	4d	4f
	spherical			

Answer is 3p, 4p & 4d subshells

which has total 3 + 3 + 5 = 11 orbitals

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2. (4.00)

Sol. $r = \frac{n^2}{Z} a_0$

for 1s $n = 1 \Rightarrow r = \frac{a_0}{z}$

but $\frac{a_0}{z} \times 4 = a_0 \Rightarrow z = 4$

3. (4.00)

Sol. $n = 7 \rightarrow n = 1$

number of spectral lines in

Paschen series

$$\underbrace{\begin{matrix} 7 & 6 & 5 & 4 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 3 & 3 & 3 & 3 \end{matrix}}_{4 \text{ lines}}$$

SECTION-IV : (Matrix Type)

1. (A) - R ; (B) - Q ; (C) - S, T ; (D) - P

Match the column

(A) $\ell \rightarrow$ possible values from 0 to $(n-1)$

(B) $s \rightarrow$ It can have only two possible values $s = +\frac{1}{2}, -\frac{1}{2}$

(C) $m \rightarrow$ Indicates the possible orientation in space possible values are $+\ell$ to $-\ell$ including zero

(D) n Indicates the distance from nuclaus

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